CS496 AI ENGINEERING · PROJECT 1 · 7 OCT 2026

The SQL Grader: where we are

Can an LLM judge reliably grade plain-language explanations of SQL queries?

Progress: 6 of 9 steps done

1–4 Queries, candidates, labels, golden set — Done	5 Judge vs labels (dev) — Done	6 Position bias — Done
7 Verbosity bias — Restarting	8 System prompt tuning — To do	9 Final test run + cost — To do

Key results

Golden set	161 items (85 good, 76 bad); dev 60, test 101
Labeller agreement (2 Claude instances)	157/161 (97.5%), κ = 0.947
Judge on dev (gpt-oss-120b, v2)	53/60 (88.3%), κ = 0.765; caught 29/33 bad answers
Position bias (101 test pairs)	Verdict flips on swap: 8/101 (7.9%). No slot preference (24.3% vs 22.8%)
Correct vs broken answer	Picked the correct one in both orders 30/35 (85.7%)
Judge cost	≈ $0.0005 per item, median 1.5 s

Findings

- **The judge found label errors.** Both labellers accepted "lists every customer" for inner joins, which drop customers with no orders. 13 labels were fixed before the judge saw the test split.
- **Judge weak spots:** passes clause-by-clause SQL restatements, and reads "for each" as "every".
- **Caveat:** 53% of pairwise verdicts are ties, which makes the judge look more stable than it is.

Blocker and next steps

The Groq free tier allows about 100–150 judge calls a day per account, and steps 7–9 need about 1,000, so we are splitting them across teammates' accounts. After that come the 4-page report and the postmortem, with the repo frozen before Week 5.

Questions for the instructor

- Is reporting the test score as the headline, and disclosing the dev label fixes, enough?
- Should position bias be reported on non-tie verdicts only?
- Can we split API runs across accounts, or use a smaller verbosity run?